

PIPE modified at 13:00 11 April 2025, Drawing Created at 18:14:53 11 April 2025

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FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PRE-SET ATTEN. CONTROL, 50 OHMS, BE T=100%	3"	2.1 M	S		PAC0-0002181
2	WAVE COUPLER AND ATTEN. IN COAXIAL CABLE	32"	2	S98	TH	PAC0-0002181
3	PRE-SET ATTEN. PRE-SET, 50 OHMS, 1000MA/10"	32"	2	S98	HPG3	PAC0-0004335
4	COIL, SINGLE BLOWN, 100 OHMS, 50 OHMS, 500W	301.12"	1	S98	ACS	PAC0-0001259
5	COIL, SINGLE BLOWN, 100 OHMS, 50 OHMS, 500W	301.12"	1	S98	E	PAC0-0001259
6	SLIP-ON FLANGE FOR 1/2" DIA. ATTEN. IN COAXIAL CABLE	3"	4	S98	F	PAC0-0002186

[illegible]

TIAO NO. PG-945-94-02, INSTRUMENT		INSTRUMENT			
SIZE	PIPE LENGTH	3"	1"		
NO	LENGTH (MM)	REMARKS	NO	LENGTH (MM)	REMARKS
1	271	1	NO	266	1
2	100	1	<2	46	3
3	150	1	46	576	3

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Project PROVIDENCE
Approval For Construction
SIPING MANAGER Bea Chang Won
2025-06-14

LOTTE BIOLOGICS

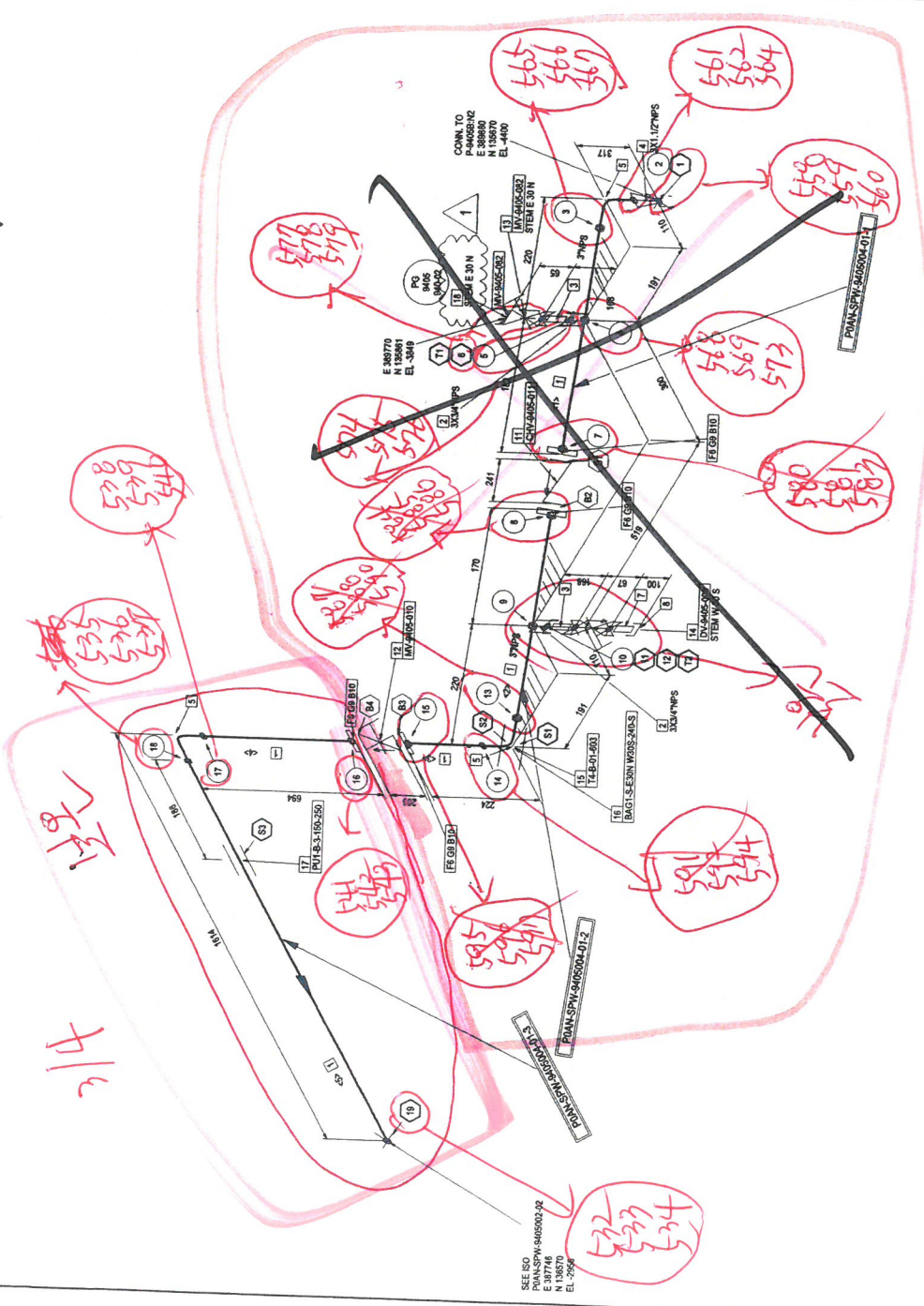


PROJECT PROVIDENCE

LOTTE E&C									
AG		PLANT		N		PDB		DMA NO.	
K1-LSL-UT-PID-a-05						10460		PWAN	
						FLUID		SPW	
NO.		NO.		AREA		PROL. NO.		LINE NO.	
NO.		NO.		NO.		NO.		NO.	
								3-SPW-9405004-SB4-5N	
								P04N-SPW-9405004-01	
								SHT NO.	
								1 OF 1	
								REV	
								1	

[illegible]

NOTE : ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

[illegible]

FABRICATION MATERIALS

EJECTION MATERIALS									
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM	NO	COMPONENT DESCRIPTION	ITEM
1	PIPE ASTM A210-10/12H, SCH10S BE RT=100%	7"	17.6M	508	P	PAC-000011903	4	PIPE EPW ASTM A210-10/12H, SCH10S BE RT=100%	PAC-000011903
2	WELDED ELBOW W/30° BEVELLED ASTM A403-HP10/12H, SCH10S	7"	3	508	E	PRB-000034110	5	O-RING	O1
3	45 DEG. ELBOW W/30° BEVELLED ASTM A403-HP10/12H, SCH10S	7"	2	508	EA	PRB-000033652	6	SP-3	SP3
CUT PIPE LENGTH									
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM	NO	COMPONENT DESCRIPTION	ITEM
1	PIPE	7"	14.6M	508	P	PAC-000011903	1	PIPE	PAC-000011903
2	WELDED ELBOW	7"	3	508	E	PRB-000034110	2	WELDED ELBOW	PRB-000034110
3	45 DEG. ELBOW	7"	2	508	EA	PRB-000033652	3	45 DEG. ELBOW	PRB-000033652
CUT PIPE LENGTH									
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM	NO	COMPONENT DESCRIPTION	ITEM
1	PIPE	7"	14.6M	508	P	PAC-000011903	1	PIPE	PAC-000011903
2	WELDED ELBOW	7"	3	508	E	PRB-000034110	2	WELDED ELBOW	PRB-000034110
3	45 DEG. ELBOW	7"	2	508	EA	PRB-000033652	3	45 DEG. ELBOW	PRB-000033652

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Project PROVIDENCE
Approval For Construction
SIPING MANAGER Bea Chang Won
2025-09-04

LOTTE BIOLOGICS



PROJECT PROVIDENCE

T	AG	PLANT	N
1	1	1	1

NO.	.
NO.	P1RN-SPW-9405002-01

SHOT NO. 1 OF 1 REV. 1

[illegible][illegible]

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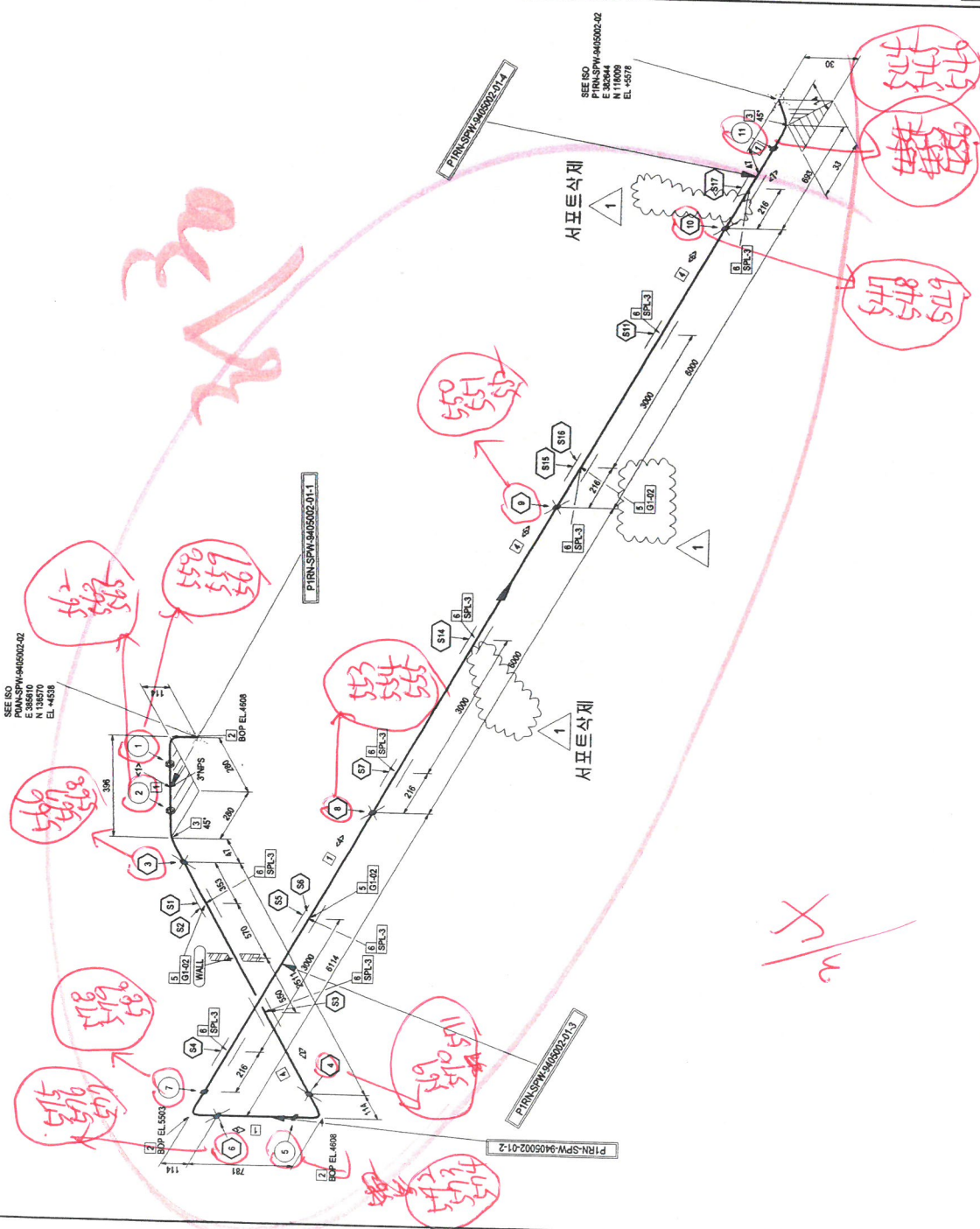
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BILL OF MATERIALS

FABRICATION MATERIALS						
NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE EPM ASTM A133/A133M SCH40S BE RP 100%	3" 108	1	SSE		PMA-200001903
2	45 DEG. ELBOW RW WELDED ASTM A403HP151818 SCH40S	3" 7.31	1	SSE	EA	P55-000003552
3	90 DEG. ELBOW LRW WELDED ASTM A403HP151818 SCH40S	3" 7"	3	SSE	E	P55-000003549

ERECTOR MATERIALS						
NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
4	DAT	3"	1	DAT		
5	0-142	3"	1		01	
6	SPLJ	3"	2	SPL		

CUT PIPE LENGTH						
NO.	PRICE	LENGTH (IN)	REMARKS	WPD (IN)	REMARKS	REMARKS
1	372	66	3	WPD (IN)		
2	42	3	3	WPD (IN)		
3	46	3	3	WPD (IN)		
4	46	3	3	WPD (IN)		
5	46	3	3	WPD (IN)		
6	46	3	3	WPD (IN)		
7	46	3	3	WPD (IN)		
8	46	3	3	WPD (IN)		
9	46	3	3	WPD (IN)		
10	46	3	3	WPD (IN)		
11	46	3	3	WPD (IN)		
12	46	3	3	WPD (IN)		
13	46	3	3	WPD (IN)		
14	46	3	3	WPD (IN)		
15	46	3	3	WPD (IN)		
16	46	3	3	WPD (IN)		
17	46	3	3	WPD (IN)		
18	46	3	3	WPD (IN)		
19	46	3	3	WPD (IN)		
20	46	3	3	WPD (IN)		
21	46	3	3	WPD (IN)		
22	46	3	3	WPD (IN)		
23	46	3	3	WPD (IN)		
24	46	3	3	WPD (IN)		
25	46	3	3	WPD (IN)		
26	46	3	3	WPD (IN)		
27	46	3	3	WPD (IN)		
28	46	3	3	WPD (IN)		
29	46	3	3	WPD (IN)		
30	46	3	3	WPD (IN)		
31	46	3	3	WPD (IN)		
32	46	3	3	WPD (IN)		
33	46	3	3	WPD (IN)		
34	46	3	3	WPD (IN)		
35	46	3	3	WPD (IN)		
36	46	3	3	WPD (IN)		
37	46	3	3	WPD (IN)		
38	46	3	3	WPD (IN)		
39	46	3	3	WPD (IN)		
40	46	3	3	WPD (IN)		
41	46	3	3	WPD (IN)		
42	46	3	3	WPD (IN)		
43	46	3	3	WPD (IN)		
44	46	3	3	WPD (IN)		
45	46	3	3	WPD (IN)		
46	46	3	3	WPD (IN)		
47	46	3	3	WPD (IN)		
48	46	3	3	WPD (IN)		
49	46	3	3	WPD (IN)		
50	46	3	3	WPD (IN)		
51	46	3	3	WPD (IN)		
52	46	3	3	WPD (IN)		
53	46	3	3	WPD (IN)		
54	46	3	3	WPD (IN)		
55	46	3	3	WPD (IN)		
56	46	3	3	WPD (IN)		
57	46	3	3	WPD (IN)		
58	46	3	3	WPD (IN)		
59	46	3	3	WPD (IN)		
60	46	3	3	WPD (IN)		
61	46	3	3	WPD (IN)		
62	46	3	3	WPD (IN)		
63	46	3	3	WPD (IN)		
64	46	3	3	WPD (IN)		
65	46	3	3	WPD (IN)		
66	46	3	3	WPD (IN)		
67	46	3	3	WPD (IN)		
68	46	3	3	WPD (IN)		
69	46	3	3	WPD (IN)		
70	46	3	3	WPD (IN)		
71	46	3	3	WPD (IN)		
72	46	3	3	WPD (IN)		
73	46	3	3	WPD (IN)		
74	46	3	3	WPD (IN)		
75	46	3	3	WPD (IN)		
76	46	3	3	WPD (IN)		
77	46	3	3	WPD (IN)		
78	46	3	3	WPD (IN)		
79	46	3	3	WPD (IN)		
80	46	3	3	WPD (IN)		
81	46	3	3	WPD (IN)		
82	46	3	3	WPD (IN)		
83	46	3	3	WPD (IN)		
84	46	3	3	WPD (IN)		

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Project PROVIDENCE

Approval For Construction
PIPING MANAGER Bea Chang Won
2025-09-04

PROJECT PROVIDENCE	 LOTTE BIOLOGICS
	 LOTTE E&C

NO.	AG	PLANT	N	AREA	PDB	CWA NO.	PTRN
NO.	K1-D8L-UT-PID-9405			PROJ. NO.	16480	FLUID	SPW
NO.				LINE NO.	3-SPW-9405002-S98-SN		
NO.	PTRN-SPW-9405002-02						
NO.						SHT NO.	REV

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Project PROVIDENCE
Approval For Construction
SUPERVISING MANAGER, Bea Chang Won
2025-09-04

LOTTE BIOLOGICS

 LOTTE E&C

PROJECT PROVIDENCE

T	AG	PLANT
1/1/01	1/1/01	1/1/01

NO.	K1-D5L-VI+PID-9405
NO.	.

NO.	P1RN-SPW-9405002-02
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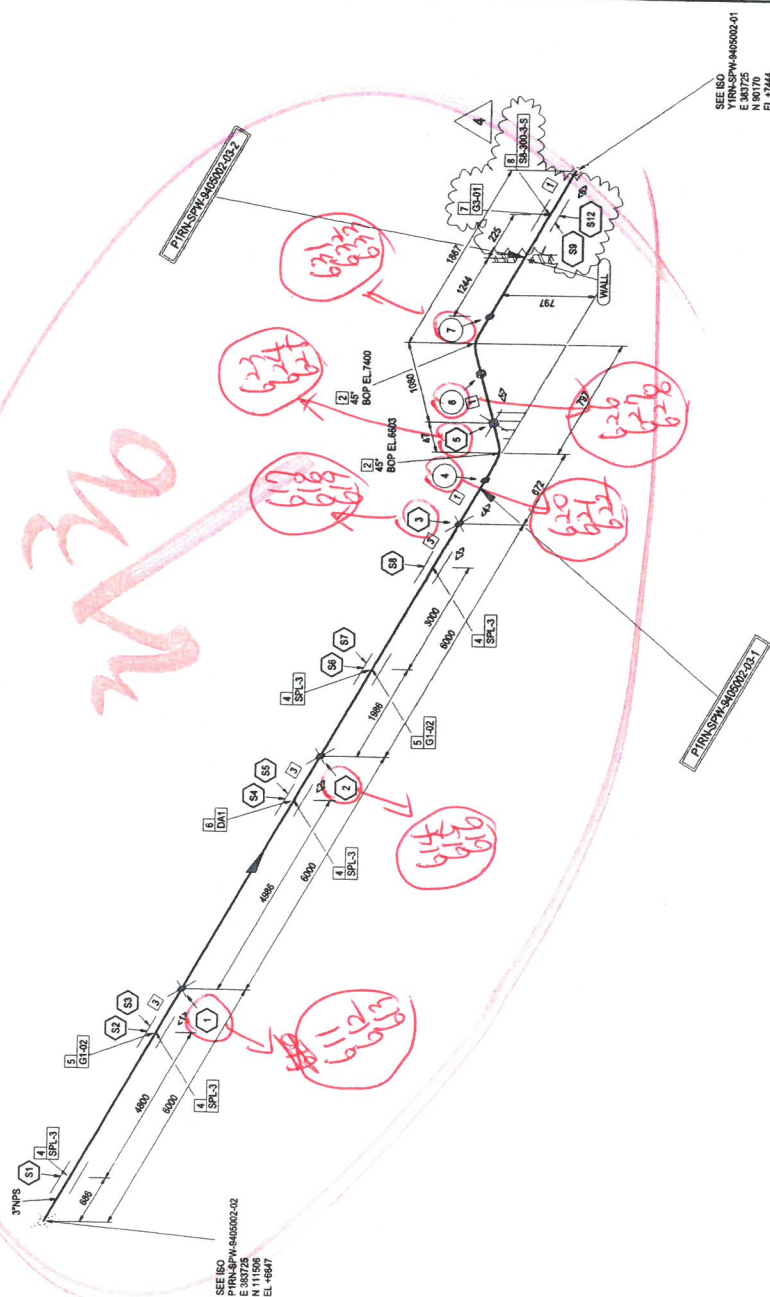
FABRICATION MATERIALS

COMPONENT DESCRIPTION						
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE EVA ASTM A1315/16/18 SCH15 BE RT=100%	3"	13.1 M	586	P	PA02-00001903
2	45 DEG ELBOW BW W/ELDED ASTM A403-WP16/18 SCH15	3"	2	586	EA	P056-00025302

ERECTOR MATERIALS						
COMPONENT DESCRIPTION						
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
3	PPE EVA ASTM A1315/16/18 SCH15 BE RT=100%	3"	13.1 M	586	P	PA02-00001903
4	SP-3	3"	1			
5	0-1/2"	3"	2			
6	DAT	3"	1			
7	03-41	3"	1			
8	58-300-5-5	3"	1			

ERECTOR LENGTH					
PREC	LENGTH	NPD	REMARKS	PREC	LENGTH
	(MM)	(IN)			(IN)
3	800	3		3	31
4	800	3		4	31
5	1000	3		4	31
6	1000	3		4	31
7	1000	3		4	31
8	1000	3		4	31

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2026-01-15

LOTTE BIOLOGICS

PROJECT PROVIDENCE

PART NO.	AG	PLANT	N	AREA	PDB	CWA NO.	PURN
2&D NO.	K1-DLSLT-PID-9405			16400	FLUID		SPW
3&D NO.	-						
LINE NO.					3-SPW-9405002-538-15N		
INSTR. NO.	P1RN-SPW-9405002-03						
						SHT NO.	REV
						1	4

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